EFECTIS ERA AVRASYA
TEST VE BELGELENDİRME A.Ş.

Fire Test Laboratory

AB-0556-T
EEA-24-010
04-24

CLASSIFICATION OF FIRE RESISTANCE PERFORMANCE IN ACCORDANCE WITH EN 13501-2:2016

Sponsor	**:** KİMFOR İZOLASYON YAPI KİMYASALLARI İNŞ. MAD. GIDA TİC. SAN. LTD. ŞTİ. Yunuseli Mah. Akyıldız Cad. No:3, Osmangazi, Bursa / TÜRKİYE
Prepared by	**:** EFECTIS ERA AVRASYA TEST VE BELGELENDİRME A.Ş. Dilovası OSB 5. Kısım Fırat Cad. No: 18, 41455 Dilovası, Kocaeli / TÜRKİYE
Product name	**:** Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"**
Classification report No.	**:** EEA-24-010
Issue number	**:** 1/2
Date of issue	**:** 04.04.2024

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1. INTRODUCTION

This classification report defines the classification in accordance with the procedures given in EN 13501-2:2016, assigned to Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"**.

2. DETAILS OF CLASSIFIED PRODUCT

2.1. General

The element Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"** is defined as a type of product.

2.2. Description

Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"** is fully described below.

2.2.1. General

Product identification : Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"**

Fire Side : From Surface Coat Side

Manufacturer : KİMFOR İZOLASYON YAPI KİMYASALLARI İNŞ. MAD. GIDA TİC. SAN. LTD. ŞTİ.
Yunuseli Mah. Akyıldız Cad. No:3, Osmangazi, Bursa / TÜRKİYE

Sponsor of test : KİMFOR İZOLASYON YAPI KİMYASALLARI İNŞ. MAD. GIDA TİC. SAN. LTD. ŞTİ.
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2.2.2. Construction

Asymmetrical, non-loadbearing wall system was comprised 5-6 lines aerated concrete blocks at horizontal axis and 12 lines aerated concrete blocks at vertical axis. Cement based mortar was used to connect the blocks. A1 Insulated Stucco Filling "THERMOLOCK" was used at exposed side of the wall as surface coat. The total thickness of the wall was 180 mm.

One vertical edge was mounted as in practice and the other vertical edge was constructed as a free edge to simulate a wider wall construction.

2.2.3. Components

2.2.3.1. Blocks

• Type : Aerated Concrete Blocks – TÜRK YTONG SANAYİ A.Ş

• Fire class : A1 (non-flammable)

• Nominal dimensions : 600 x 150 x 250 mm (w x d x h)

• Compressive strength : 2,5 N/mm2

• Nominal gross dry density: 400 kg/m3

2.2.3.2. Mortar

• Type : Cement based adhesive mortar – YTONG Örgü Tutkalı

• Fire class : A1 (non-flammable)

• Location : Used between the blocks to connect.

• Nominal density : 1400 kg/m3

• Nominal thickness (Joint gap): 2 mm

2.2.3.3. Surface Coat

• Type : Cement based A1 Insulated Stucco Filling "THERMOLOCK"

• Fire class: A1 (non-flammable)

• Location: At exposed side of the wall

• Nominal dry density: 333 kg/m3

• Measured moisture content: 7,77 %

• Nominal thickness: 30 mm

See figure 1 – figure 4 for detailed information**.**

3. REPORTS AND RESULTS IN SUPPORT OF CLASSIFICATION

3.1. Reports

Name of laboratory Name of sponsor Test report ref. no.			Test method
EFECTIS ERA AVRASYA TEST VE BELGELENDİRME A.Ş.	KİMFOR İZOLASYON YAPI KİMYASALLARI İNŞ. MAD. GIDA TİC. SAN. LTD. ŞTİ.	RFTR24028	EN 1364-1:2015

3.2. Results

Test Method	Parameter	Results
EN 1364-1:2015	Integrity, [**E**]	
	– Cotton pad	No failure (not applied).
	– Gap gauges Ø 6 mm	No failure (not applied).
	Ø 25 mm	No failure (not applied).
	– Flames longer than 10 s	Not observed.
	Insulation, [**I**]	
	– Average temperature	No failure.
	– Maximum temperature	No failure.
Test was terminated at the 241st minute after consulted with sponsor.		

4. CLASSIFICATION AND FIELD OF APPLICATION

4.1. Reference of classification

This classification has been carried out in accordance with clause 7.5.2 of EN 13501-2:2016.

4.2. Classification

Non-loadbearing Wall Consisted of Aerated Concrete Blocks and A1 Insulated Stucco Filling **"THERMOLOCK"** is classified according to the following combinations of performance parameters and classes as appropriate.

R	E	I	W		t	t	-	M	S	C	IncSlow	sn	ef	r

FIRE RESISTANCE CLASSIFICATION	
Fire Side: Surface Coat Side	**E240, EI240**

4.3. Field of application

4.3.1. General

This report details the method of construction, the test conditions and the results obtained when the specific elements of construction described herein was tested following the procedure outlined in EN 1363-1:2020, and when appropriate EN 1363-2:1999. Any significant deviation with respect to size, constructional details, load stresses, edge or end conditions other than those allowed under the field of direct application in the relevant test method is not covered by this report.

4.3.2. Specific conditions for dimensions and fixing

Test results are directly applicable to similar constructions where one or more of the changes listed below are made and the construction continues to comply with the appropriate design code for its stiffness and stability.

- decrease in height
- increase in the thickness of the wall
- increase in the thickness of component materials

4.3.3. Extension of height

The height of the construction may be increased up to 1 meter for a fire resistance of **240 minutes** for the criteria of thermal insulation and integrity, because the lateral deflection of the construction is less than 100 mm.

4.3.4. Extension of width

The width of an identical construction may be increased, because the construction is tested with a width of 3 meters with one vertical free edge.

5. LIMITATIONS

This classification report does not represent any type approval or certification of the product. This report is initially valid until **4th April 2029** providing that no significant modifications are made in technical specification of the specimen and related test and classification standards.

Signed:	Approved:
e-signed	e-signed
Yusuf ÜSTÜNDAĞ	Ali BAYRAKTAR
Person in charge of test	Laboratory Manager

6. DRAWINGS

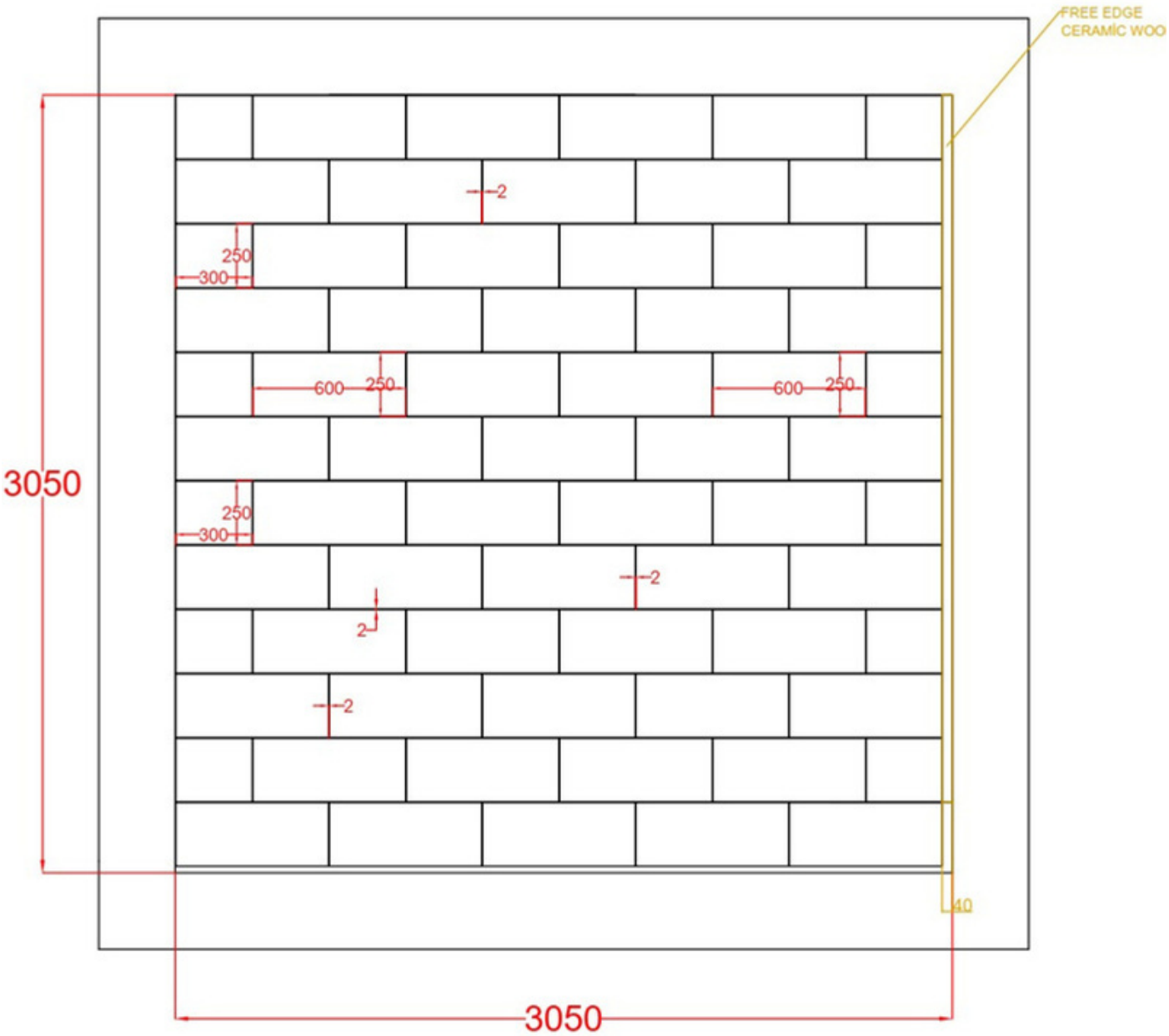

Figure 1: Unexposed side view of the test specimen.

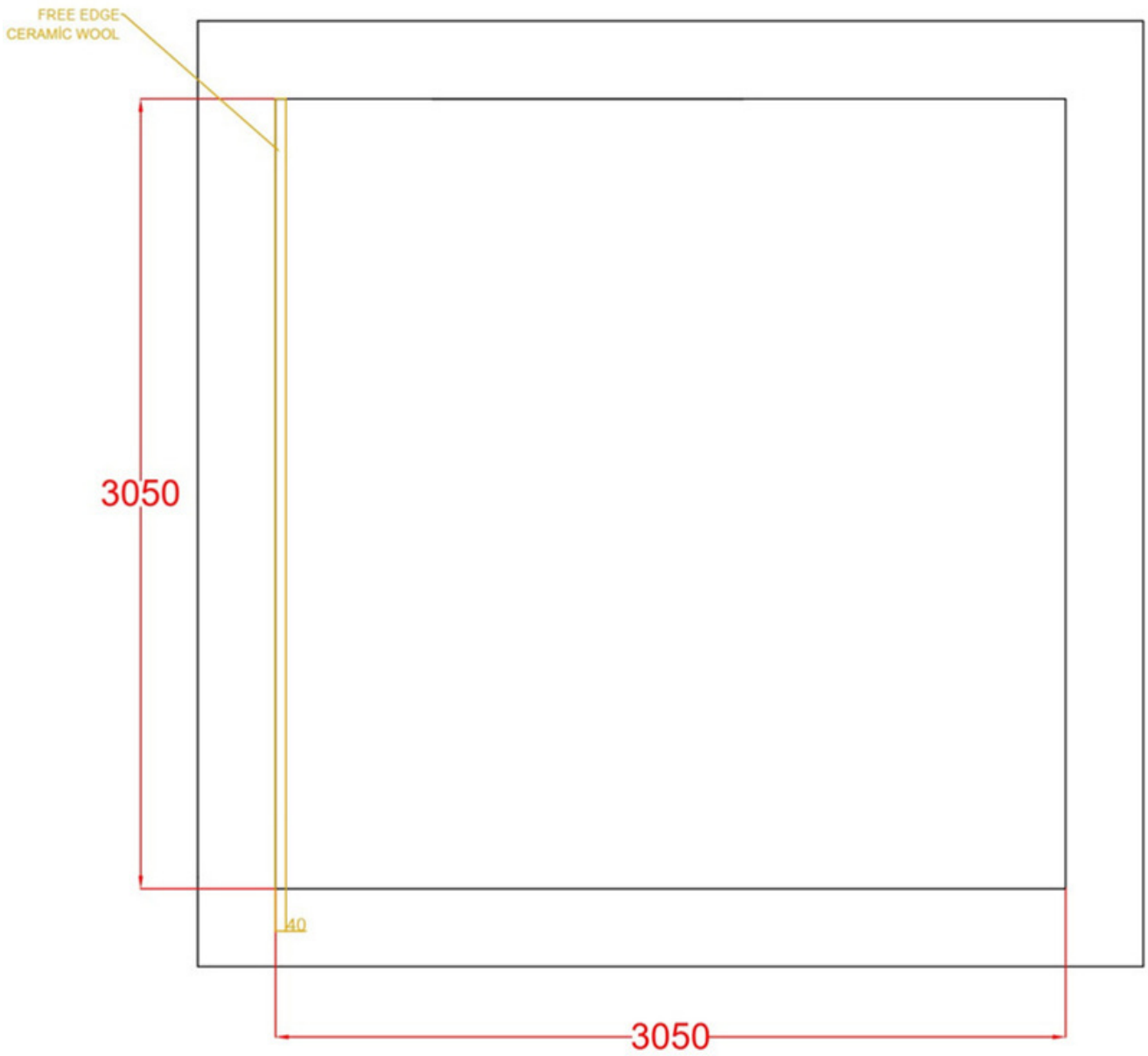

Figure 2: Exposed side view of the test specimen.

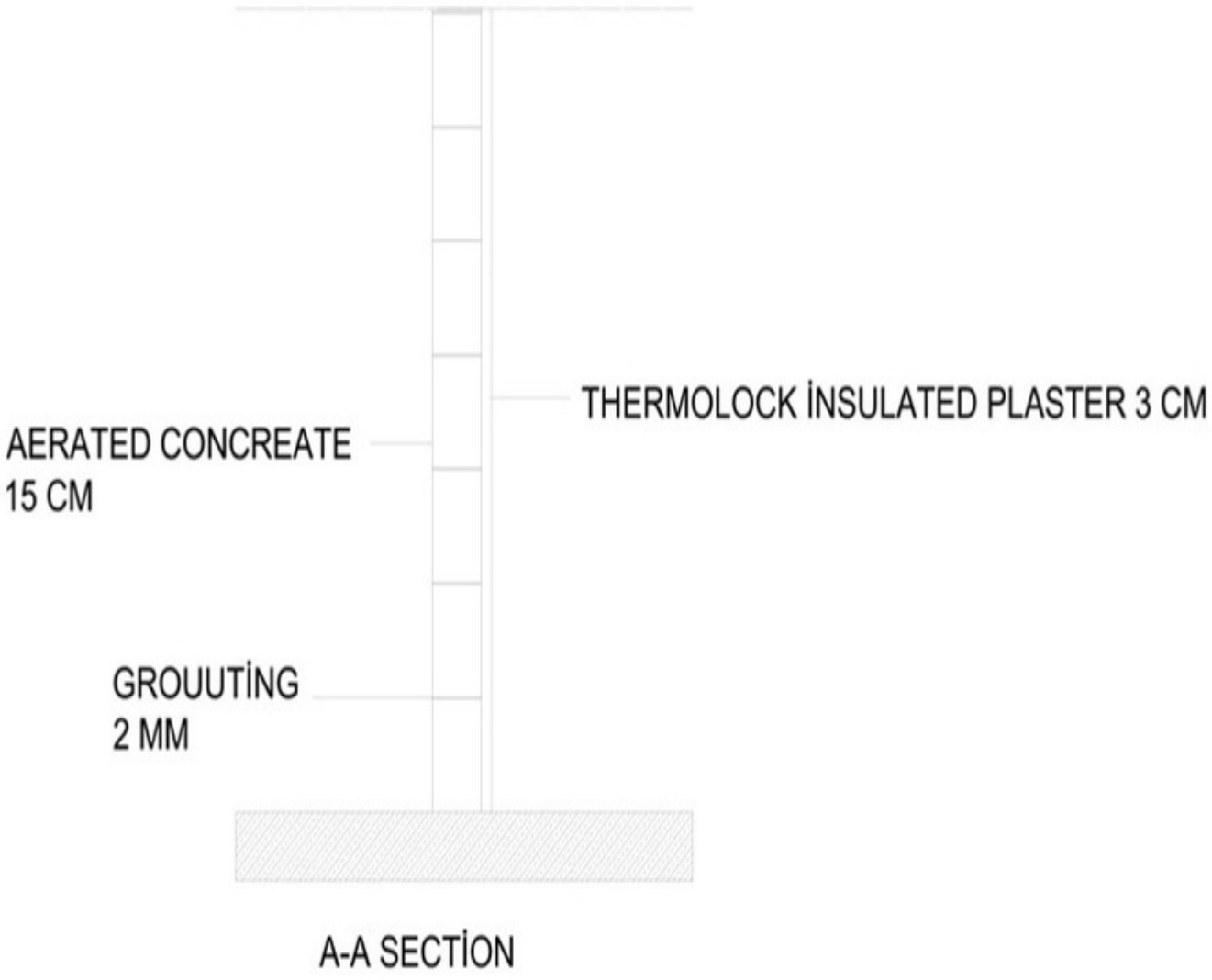

Figure 3: Longitudinal view of the test specimen

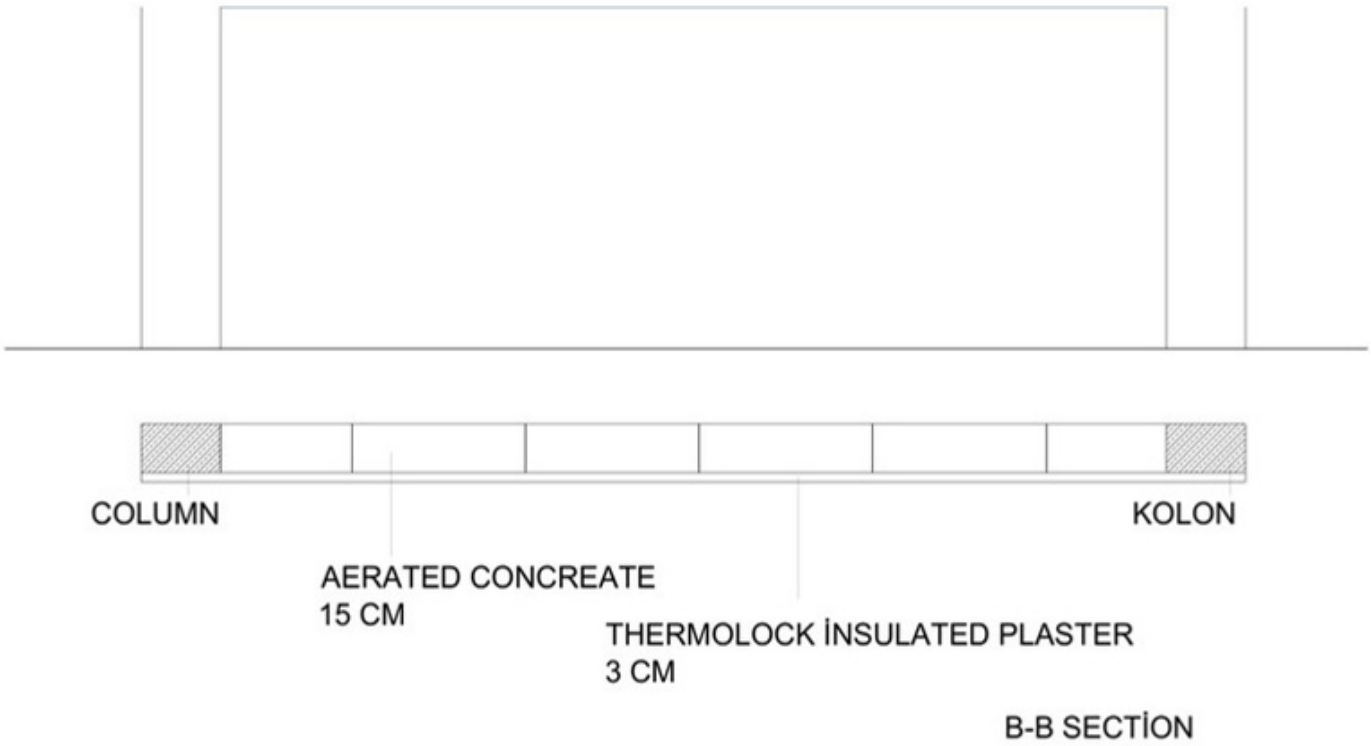

Figure 4: Cross section view of the test specimen.